The Pharma Innovation Journal 2023; 12(12): 4048-4051

www.ThePharmaJournal.com

The Pharma Innovation

ISSN (E): 2277-7695
ISSN (P): 2349-8242
NAAS Rating: 5.23
TPI 2023; 12(12): 4048-4051
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www.thepharmajournal.com
Received: 01-10-2023
Accepted: 04-11-2023

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Detection of potato leaf disease with application of machine learning techniques

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Abstract
Potato is one of the most widely consumed crop in India, also there is significant very high demand of potatoes globally. However, some potato diseases are the leading cause of declination of potato quality and production. Diseases, such as early blight, late blight, common scab, potato leafroll virus etc. have a significant impact on the production quality and quantity, an automated system based on computer vision can help the farmers for early detection of leaf diseases. This study proposes a machine learning based approach for early detection of potato leaf diseases. Overall 2150 samples have been observed from online database. The overall samples are divided into three class: early blight, late blight and healthy and divided into three groups (Group1: early blight and healthy, Group2: late blight and healthy and Group 3: early blight and late blight). Various machine learning algorithms has been employed for classification of early blight, late blight and healthy potato leafs. An highest accuracy of 98.4% has been achieved using convolution neural network (CNN) for late blight nad healthy group, indicating the feasibility of the machine learning techniques and may help to reduce losses incurred by the potato farmers due to crop diseases.

Keywords: Potato leaf disease, machine learning, CNN

Introduction
The Indian country economy is heavily dependent on agriculture; potatoes are one of the most widely consumed crops across India. Potatoes are rich in various nutrients and can be cooked with various other vegetables making it popular among Indian household [1]. The crop diseases may affect the plant including leafs, steams, roots which may cause loss in production. At present, crop disease detection is mainly done by visual inspection by plant pathologists that requires lot of time and may report inaccurate results, in order to minimize losses caused due to crop diseases, there is requirement of automation in the field of agriculture for detection of plant diseases [2]. There is also problem that different parts of country may effect by different diseases. There are various studies going on for early detection of plant diseases by applying computer vision and machine learning [3]. The potato leaf diseases may be classified using various image processing techniques that employs machine learning to classify diseased leaf with healthy one [2-4]. The machine learning algorithm works on identifying visual symptoms that includes discoloration, spots etc, through various image processing techniques thus identifying the characteristics of particular diseases. Among the various machine learning techniques available, convolutional neural network (CNN) shows promising results for identification of plant leaf diseases [5-9]. In our study we uses CNN model to classify potato leaf infected by disease and healthy leaves. The potato diseases included in our study are early blight and late blight. The proposed approach can help the farmers for early detection of crop leaf disease thus, improving the production quality and quantity.

Materials and Methods
The block diagram shows proposed methodology for our study is shown in figure 1. The steps include data acquisition, pre-processing, building CNN model, classification of diseased leaf with health ones.

Data Collection: The data of potato leaves infected from early blight and late blight diseases were taken from kaggle plant village database available publically. Overall 2150 samples has been taken. The figure 2, 3 and 4 shows the potato leaves infected from early blight, late blight and healthy leaves.

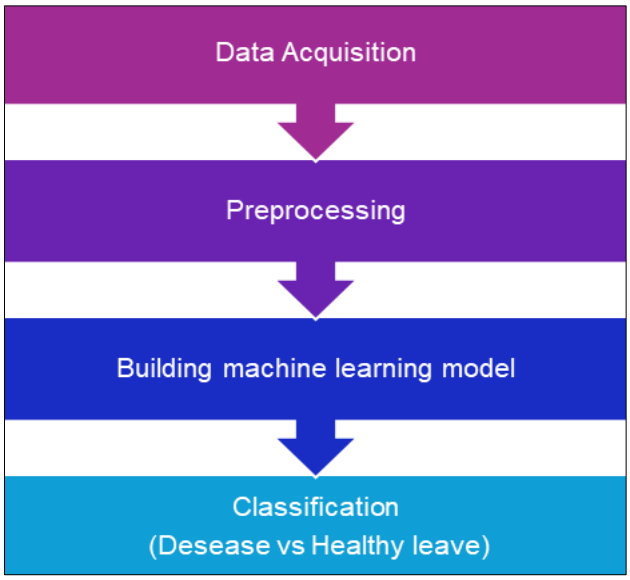

Fig 1: Block diagram of early detection of leaf disease.

Fig 2: Potato leaves infected by early blight.

Fig 3: Potato leaves infected by late blight.

Fig 4: Potato healthy leaves.

Pre-processing: The first step in preprocessing stage is to resize each and every image in 256 x 256 pixel format. The next step is to remove noise from the image by applying basic filtering techniques and after that normalization of pixel value is done in the range 0-1.

Machine learning model: The classifier learner application in MATLABR2018a is used for classification with 10 fold cross validation. The machine learning algorithms used for classification includes: Support vector machine (SVM), CNN, K- nearest neighbor (KNN) and tree. The classification results of all the classifiers are compared on the basis of accuracy.

Classification: The images are divided into three classes: early blight, late blight and healthy group. The classification was done in three groups. In first group classification was done among early blight and healthy group, the second group classification was done among late blight and healthy group and in last group classification was done among early blight and late blight group. The classification results of all the machine learning algorithms are then compared on the basis of their accuracy.

Results

In our study we have successfully classified infected potato leaves (early blight and late blight) with healthy one. For doing this classification we used image processing by employing machine learning algorithms (CNN, KNN, SVM, Tree). The machine learning algorithms was compared on basis of their accuracy. The comparison results for the three groups discussed in the materials and method section are shown in Table 1 and figure 5.

Table 1: Comparison results of machine learning on the basis accuracy for all three groups.

S. No	Machine learning algorithms	Accuracy (in %) Early blight vs Healthy group)	Accuracy (in %) Late blight vs Healthy group)	Accuracy (in %) Early blight vs Late blight)
1	CNN	94.7	98.4	89.6
2	SVM	92.6	95.1	87.4
3	KNN	88.4	89.4	90.5
4	Tree	90.6	91.8	87.8

The result shows that CNN is having highest accuracy of 94.7% when classifying between early blight an healthy group, whereas highest accuracy of 98.4% is obtained while classifying between late blight and healthy group using CNN model. The least accuracy was reported for early blight and late blight group, were KNN shows the accuracy of 90.5%.

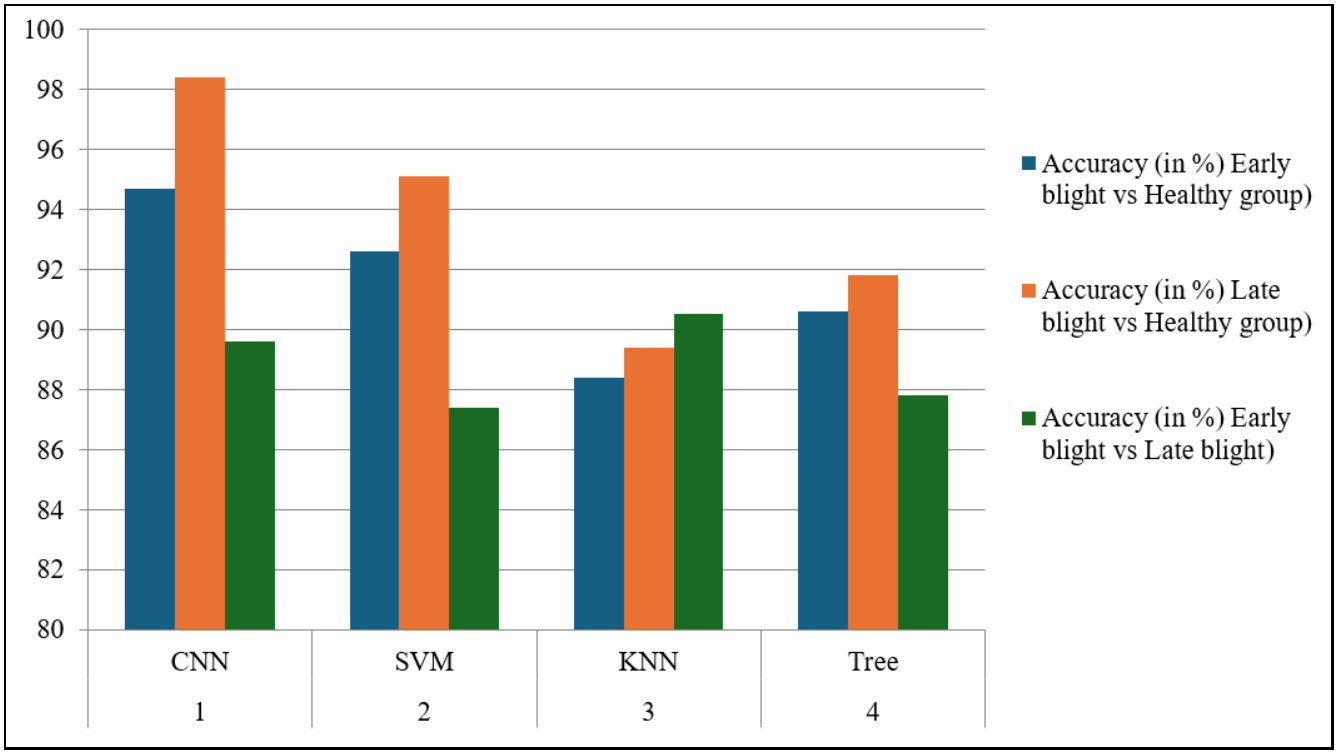

Fig 5: Comparison graph of machine learning on the basis accuracy for all three groups

Although all the algorithms we have considered performed very well but, we have got appreciable results for all the groups with highest accuracy 98.4% was observed while classifying between late blight and healthy group. The result shows the effectiveness of performance of various machine learning algorithms.

Conclusion

In this study we have divided potato leaves in three classes: early blight, late blight and healthy. After that divided into three groups (Group1: early blight and healthy, Group2: late blight and healthy and Group 3: early blight and late blight). Various machine learning was applied for classification and compared on the basis of their accuracy, the best result with accuracy of 98.4% was obtained while classifying late blight and healthy with CNN model.

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